

MLFF PERF-P1 CPU Qualification

Exact shared selection and progressive-coverage benchmark

mdstats project

2026-08-15

1 Scope

This report measures `mdstats 0.20.180a0` PERF-P1 against embedded pre-P1 exact oracles. It uses the authenticated PERF-P0 TARGET-DATA2B native reference rather than re-parsing VASP XML, so the selector/coverage input is identical while XML I/O is excluded from this gate-local timing.

2 Host

Property	Value
CPU	Intel Xeon Platinum 8573C
Visible logical CPUs	9
cgroup CPU quota	8 cores (800000 100000)
cgroup memory limit	4 GiB
Python	3.13.5
OMP/OpenBLAS threads	8 / 8

3 Scientific equivalence

Check	Result
Full fused selector matrix	Exact
Full exact FPS order through 1024	Exact
Coverage reports at 128/256/512/1024	Exact
Wide 4000x128 FPS order through 512	Exact
DATA7 K=8192 nearest-selected minima	Exact

Reference content digest:

4f46dfaa6c366ede1cd4d19cf5fb8be65cfe49067b5aff85d668e0078915f9b8

Scientific evidence digest:

ff08ca4aee884f1aaf4bf1969454bb75fc9e875eb8c29d57c46fe0100dadb12e

4 Full-reference matched measurements

Five repetitions were recorded for each path over 37,633 frames and a 50-column selector matrix.

Stage	Legacy median	PERF-P1 median	Relative change
Selector assembly wall	0.1039 s	0.1050 s	+0.98%
FPS wall, K=1024	1.5503 s	0.6571 s	-57.61%
Four-rung coverage wall	0.6425 s	0.7086 s	+10.29%
Assembly RSS increment	30.25 MiB	0.008 MiB	allocator-sensitive reduction

FPS is materially faster because row norms, selected membership/ranks, and minimum squared distances persist in one workspace instead of being rebuilt. Selector assembly is timing-neutral but removes the final concatenation allocation. Progressive coverage is exact but slower for this four-rung case; that result is retained as measured rather than normalized away.

5 Wide exact FPS

A deterministic synthetic 4000x128 selector matrix isolates the wide-feature case.

Path	Wall
Legacy	0.1821 s
PERF-P1	0.0487 s

The PERF-P1 path is **73.25% faster** with identical ordered selection.

6 DATA7 K-large memory case

A separate child process measures selected-neighbor coverage for K=8192, D=16, with rungs 1024/2048/4096/8192.

Metric	Dense legacy	PERF-P1	Change
Persistent neighbor state	512 MiB	64 KiB	8192x smaller
Wall	1.9535 s	0.5061 s	-74.09%
Peak RSS	1149.89 MiB	637.22 MiB	-44.58%

The final minimum-distance SHA-256 is identical:

8a6157004024d3c0964e3ca129dd884a555314209f6441d726ec02e0b99781ac.

7 Decision

PERF-P1 satisfies its bounded acceptance gate. Exact deterministic content is unchanged while the dominant exact FPS path and K-large DATA7 path show material wall/memory improvements. Progressive coverage reuse is retained for shared-state architecture and exact worker-budget control, but its observed four-rung slowdown remains an optimization target rather than a claimed speedup.

Machine-readable evidence:

audits/analysis/mlff_perf_p1_lta_cloud_cpu_2026-08-15.json.